

Stage 1: Cleaning and Preprocessing Planet Data

```
In [1]: # This code was created by me using Lecture videos from the Data Science course
#and Database and Advanced Data Techniques and previous projects in Data Science Li
## my code starts here

#Load Libraries
import pandas as pd
import numpy as np
```

Planet Column Data

```
In [2]: #Load full column data file
column_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced Te

#check first rows to verify it loaded correctly
print(column_data.head(1))
```

```
   column_id abbrev_name  full_name
0          1  planet_id  Planet ID
```

```
In [3]: ##### Find critical columns in column data
critical_column_data=['pl_name', 'hostname', 'pl_letter', 'sy_snum', 'sy_pnum',
                      'sy_mnum', 'discoverymethod', 'disc_year', 'disc_facility',
                      'pl_orbper', 'pl_orbsmax', 'pl_angsep', 'pl_rade', 'pl_bma
                      'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                      'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                      'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                      'st_radv', 'ra', 'dec', 'sy_dist']

new_column_data = column_data[column_data['abbrev_name'].isin(critical_column_data)]

#new_column_data = column_data[critical_column_data]
print(new_column_data)
```

	column_id	abbrev_name	full_name
1	2	pl_name	Planet Name
2	3	hostname	Host Name
3	4	pl_letter	Planet Letter
9	10	sy_snum	Number of Stars
10	11	sy_pnum	Number of Planets
11	12	sy_mnum	Number of Moons
13	14	discoverymethod	Discovery Method
14	15	disc_year	Discovery Year
18	19	disc_facility	Discovery Facility
32	33	pl_orbper	Orbital Period [days]
37	38	pl_orbsmax	Orbit Semi-Major Axis [au]
42	43	pl_angsep	Angular Separation [mas]
47	48	pl_rade	Planet Radius [Earth Radius]
57	58	pl_bmasse	Planet Mass or Mass*sin(i) [Earth Mass]
68	69	pl_dens	Planet Density [g/cm**3]
73	74	pl_orbeccen	Eccentricity
78	79	pl_insol	Insolation Flux [Earth Flux]
83	84	pl_eqt	Equilibrium Temperature [K]
88	89	pl_orbincl	Inclination [deg]
93	94	pl_tranmid	Transit Midpoint [days]
105	106	pl_trandep	Transit Depth [%]
110	111	pl_trandur	Transit Duration [hours]
141	142	pl_rvamp	Radial Velocity Amplitude [m/s]
156	157	st_spectype	Spectral Type
163	164	st_rad	Stellar Radius [Solar Radius]
168	169	st_mass	Stellar Mass [Solar mass]
173	174	st_met	Stellar Metallicity [dex]
179	180	st_lum	Stellar Luminosity [log(Solar)]
184	185	st_logg	Stellar Surface Gravity [log10(cm/s**2)]
189	190	st_age	Stellar Age [Gyr]
194	195	st_dens	Stellar Density [g/cm**3]
199	200	st_vsin	Stellar Rotational Velocity [km/s]
204	205	st_rotp	Stellar Rotational Period [days]
209	210	st_radv	Systemic Radial Velocity [km/s]
215	216	ra	RA [deg]
217	218	dec	Dec [deg]
233	234	sy_dist	Distance [pc]

```
In [4]: #redoing the internal index column, 0 to 37
new_column_data=new_column_data.reset_index(drop=True)
print(new_column_data)
```

	column_id	abbrev_name	full_name
0	2	pl_name	Planet Name
1	3	hostname	Host Name
2	4	pl_letter	Planet Letter
3	10	sy_snum	Number of Stars
4	11	sy_pnum	Number of Planets
5	12	sy_mnum	Number of Moons
6	14	discoverymethod	Discovery Method
7	15	disc_year	Discovery Year
8	19	disc_facility	Discovery Facility
9	33	pl_orbper	Orbital Period [days]
10	38	pl_orbsmax	Orbit Semi-Major Axis [au]
11	43	pl_angsep	Angular Separation [mas]
12	48	pl_rade	Planet Radius [Earth Radius]
13	58	pl_bmasse	Planet Mass or Mass*sin(i) [Earth Mass]
14	69	pl_dens	Planet Density [g/cm**3]
15	74	pl_orbeccen	Eccentricity
16	79	pl_insol	Insolation Flux [Earth Flux]
17	84	pl_eqt	Equilibrium Temperature [K]
18	89	pl_orbinc1	Inclination [deg]
19	94	pl_tranmid	Transit Midpoint [days]
20	106	pl_trandep	Transit Depth [%]
21	111	pl_trandur	Transit Duration [hours]
22	142	pl_rvamp	Radial Velocity Amplitude [m/s]
23	157	st_spectype	Spectral Type
24	164	st_rad	Stellar Radius [Solar Radius]
25	169	st_mass	Stellar Mass [Solar mass]
26	174	st_met	Stellar Metallicity [dex]
27	180	st_lum	Stellar Luminosity [log(Solar)]
28	185	st_logg	Stellar Surface Gravity [log10(cm/s**2)]
29	190	st_age	Stellar Age [Gyr]
30	195	st_dens	Stellar Density [g/cm**3]
31	200	st_vsin	Stellar Rotational Velocity [km/s]
32	205	st_rotp	Stellar Rotational Period [days]
33	210	st_radv	Systemic Radial Velocity [km/s]
34	216	ra	RA [deg]
35	218	dec	Dec [deg]
36	234	sy_dist	Distance [pc]

```
In [5]: #redoing the planet's column id, 1 to length of column, 1 to 38
new_column_data['column_id']=range(1, len(new_column_data) + 1)
print(new_column_data)
```

	column_id	abbrev_name	full_name
0	1	pl_name	Planet Name
1	2	hostname	Host Name
2	3	pl_letter	Planet Letter
3	4	sy_snum	Number of Stars
4	5	sy_pnum	Number of Planets
5	6	sy_mnum	Number of Moons
6	7	discoverymethod	Discovery Method
7	8	disc_year	Discovery Year
8	9	disc_facility	Discovery Facility
9	10	pl_orbper	Orbital Period [days]
10	11	pl_orbsmax	Orbit Semi-Major Axis [au]
11	12	pl_angsep	Angular Separation [mas]
12	13	pl_rade	Planet Radius [Earth Radius]
13	14	pl_bmasse	Planet Mass or Mass*sin(i) [Earth Mass]
14	15	pl_dens	Planet Density [g/cm**3]
15	16	pl_orbeccen	Eccentricity
16	17	pl_insol	Insolation Flux [Earth Flux]
17	18	pl_eqt	Equilibrium Temperature [K]
18	19	pl_orbinc1	Inclination [deg]
19	20	pl_tranmid	Transit Midpoint [days]
20	21	pl_trandep	Transit Depth [%]
21	22	pl_trandur	Transit Duration [hours]
22	23	pl_rvamp	Radial Velocity Amplitude [m/s]
23	24	st_spectype	Spectral Type
24	25	st_rad	Stellar Radius [Solar Radius]
25	26	st_mass	Stellar Mass [Solar mass]
26	27	st_met	Stellar Metallicity [dex]
27	28	st_lum	Stellar Luminosity [log(Solar)]
28	29	st_logg	Stellar Surface Gravity [log10(cm/s**2)]
29	30	st_age	Stellar Age [Gyr]
30	31	st_dens	Stellar Density [g/cm**3]
31	32	st_vsin	Stellar Rotational Velocity [km/s]
32	33	st_rotp	Stellar Rotational Period [days]
33	34	st_radv	Systemic Radial Velocity [km/s]
34	35	ra	RA [deg]
35	36	dec	Dec [deg]
36	37	sy_dist	Distance [pc]

```
In [6]: #data types and null values for column data
new_column_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  -
0   column_id    37 non-null     int64
1   abbrev_name  37 non-null     str
2   full_name    37 non-null     str
dtypes: int64(1), str(2)
memory usage: 1020.0 bytes
```

```
In [7]: #check for missing values or Na in each column,
#True for missing, false for NOT missing
row_missing = new_column_data.isna()
```

```
#adds the number of rows in each column with missing values
row_total=row_missing.sum()
print(row_total)
```

```
column_id      0
abbrev_name    0
full_name      0
dtype: int64
```

```
In [8]: #store cleaned data in new CSV file
cleaned_column_data = new_column_data.copy()
cleaned_column_data.to_csv('cleaned_column_data.csv', index=False)
print(cleaned_column_data)
```

	column_id	abbrev_name	full_name
0	1	pl_name	Planet Name
1	2	hostname	Host Name
2	3	pl_letter	Planet Letter
3	4	sy_snum	Number of Stars
4	5	sy_pnum	Number of Planets
5	6	sy_mnum	Number of Moons
6	7	discoverymethod	Discovery Method
7	8	disc_year	Discovery Year
8	9	disc_facility	Discovery Facility
9	10	pl_orbper	Orbital Period [days]
10	11	pl_orbsmax	Orbit Semi-Major Axis [au]
11	12	pl_angsep	Angular Separation [mas]
12	13	pl_rade	Planet Radius [Earth Radius]
13	14	pl_bmasse	Planet Mass or Mass*sin(i) [Earth Mass]
14	15	pl_dens	Planet Density [g/cm**3]
15	16	pl_orbeccen	Eccentricity
16	17	pl_insol	Insolation Flux [Earth Flux]
17	18	pl_eqt	Equilibrium Temperature [K]
18	19	pl_orbincl	Inclination [deg]
19	20	pl_tranmid	Transit Midpoint [days]
20	21	pl_trandep	Transit Depth [%]
21	22	pl_trandur	Transit Duration [hours]
22	23	pl_rvamp	Radial Velocity Amplitude [m/s]
23	24	st_spectype	Spectral Type
24	25	st_rad	Stellar Radius [Solar Radius]
25	26	st_mass	Stellar Mass [Solar mass]
26	27	st_met	Stellar Metallicity [dex]
27	28	st_lum	Stellar Luminosity [log(Solar)]
28	29	st_logg	Stellar Surface Gravity [log10(cm/s**2)]
29	30	st_age	Stellar Age [Gyr]
30	31	st_dens	Stellar Density [g/cm**3]
31	32	st_vsin	Stellar Rotational Velocity [km/s]
32	33	st_rotp	Stellar Rotational Period [days]
33	34	st_radv	Systemic Radial Velocity [km/s]
34	35	ra	RA [deg]
35	36	dec	Dec [deg]
36	37	sy_dist	Distance [pc]

Planet Data

```
In [9]: #load full planet data file
planet_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced Te

#correct discovery year to an integer, otherwise it will turn to float numbers
planet_data['disc_year']=planet_data['disc_year'].astype('Int64')

#check first rows to verify it loaded correctly
print(planet_data.head(1))
```

```
planet_id  pl_name hostname pl_letter  hd_name  hip_name  tic_id \
0          1  11 Com b    11 Com          b  HD 107383  HIP 60202  TIC 72437047

          gaia_dr2_id          gaia_dr3_id  sy_snum  ... \
0  Gaia DR2 3946945413106333696  Gaia DR3 3946945413106333696      2  ...

sy_kepmagerr1  sy_kepmagerr2 \
0          NaN          NaN

          sy_kepmag_reflink pl_nnotes  st_nphot \
0  <a refstr=STASSUN_ET_AL__2019 href=https://ui....      2      1

st_nrvcc st_nspec pl_nspec pl_ntranspec pl_ndispec
0          2          0          0          0          0
```

[1 rows x 320 columns]

C:\Users\maybe\AppData\Local\Temp\ipykernel_23232\2766728904.py:2: DtypeWarning: Columns (0: pl_imppar_reflink, 1: pl_trandep_reflink, 2: pl_trandur_reflink, 3: pl_ratr_or_reflink, 4: pl_occdep_reflink, 5: pl_projobliq_reflink, 6: pl_trueobliq_reflink) have mixed types. Specify dtype option on import or set low_memory=False.

```
planet_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced T
echniques\full_planets_data.csv")
```

```
In [10]: ### data types and null values for planet data
planet_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6158 entries, 0 to 6157
Columns: 320 entries, planet_id to pl_ndispec
dtypes: Int64(1), float64(215), int64(24), str(80)
memory usage: 15.0 MB
```

```
In [11]: ##### Find critical columns
critical_planet_columns=['pl_name', 'hostname', 'pl_letter', 'sy_snum', 'sy_pnum',
                        'sy_mnum', 'discoverymethod', 'disc_year', 'disc_facility',
                        'pl_orbper', 'pl_orbsmax', 'pl_angsep', 'pl_rade', 'pl_bma',
                        'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl',
                        'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                        'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                        'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data = planet_data[critical_planet_columns]
print(new_planet_data)
```

	pl_name	hostname	pl_letter	sy_snum	sy_pnum	sy_mnum	\
0	11 Com b	11 Com	b	2	1	0	
1	11 UMi b	11 UMi	b	1	1	0	
2	14 And b	14 And	b	1	1	0	
3	14 Her b	14 Her	b	1	2	0	
4	16 Cyg B b	16 Cyg B	b	3	1	0	
...	...	...	...	...	...	...	...
6153	ups And b	ups And	b	2	3	0	
6154	ups And c	ups And	c	2	3	0	
6155	ups And d	ups And	d	2	3	0	
6156	ups Leo b	ups Leo	b	1	1	0	
6157	xi Aql b	xi Aql	b	1	1	0	

	discoverymethod	disc_year	disc_facility	\
0	Radial Velocity	2007	Xinglong Station	
1	Radial Velocity	2009	Thueringer Landessternwarte Tautenburg	
2	Radial Velocity	2008	Okayama Astrophysical Observatory	
3	Radial Velocity	2002	W. M. Keck Observatory	
4	Radial Velocity	1996	Multiple Observatories	
...	...	...	...	...
6153	Radial Velocity	1996	Lick Observatory	
6154	Radial Velocity	1999	Multiple Observatories	
6155	Radial Velocity	1999	Multiple Observatories	
6156	Radial Velocity	2021	Okayama Astrophysical Observatory	
6157	Radial Velocity	2007	Okayama Astrophysical Observatory	

	pl_orbper	...	st_lum	st_logg	st_age	st_dens	st_vsin	st_rotp	\
0	323.210000	...	1.97823	2.450000	NaN	NaN	1.20	NaN	
1	516.219970	...	2.42951	1.930000	1.56	NaN	1.50	NaN	
2	186.760000	...	1.83992	2.550000	4.50	NaN	2.60	NaN	
3	1766.410000	...	-0.15273	4.450000	4.60	1.273928	1.00	NaN	
4	798.500000	...	0.09729	4.360000	7.40	1.011029	2.70	NaN	
...	...	...	...	...	...	...	...	...	...
6153	4.617033	...	0.52541	4.125176	5.00	0.369930	9.20	12.0	
6154	241.258000	...	0.52541	4.125176	5.00	0.369930	9.20	12.0	
6155	1276.460000	...	0.52541	4.125176	5.00	0.369930	9.20	12.0	
6156	385.200000	...	1.80003	2.460000	NaN	NaN	1.78	NaN	
6157	136.970000	...	1.76745	2.610000	7.10	NaN	NaN	NaN	

	st_radv	ra	dec	sy_dist
0	43.36898	185.178779	17.793252	93.1846
1	-17.52023	229.274595	71.823943	125.3210
2	-59.72633	352.824150	39.235837	75.4392
3	-13.82260	242.602101	43.816362	17.9323
4	-28.10000	295.465642	50.516824	21.1397
...	...	...	...	...
6153	-28.65500	24.198353	41.403815	13.4054
6154	-28.65500	24.198353	41.403815	13.4054
6155	-28.65500	24.198353	41.403815	13.4054
6156	NaN	174.237219	-0.823564	52.5973
6157	-41.66164	298.562449	8.461105	56.1858

[6158 rows x 37 columns]

```
In [12]: #check for missing values or Na in each column,
         #True for missing, false for NOT missing
```

```
planet_missing = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=planet_missing.sum()
print(row_total)
```

```
pl_name          0
hostname         0
pl_letter        0
sy_snum          0
sy_pnum          0
sy_mnum          0
discoverymethod  0
disc_year        1
disc_facility    0
pl_orbper        337
pl_orbsmax       315
pl_angsep        344
pl_rade          50
pl_bmasse        31
pl_dens          139
pl_orbeccen      938
pl_insol         1842
pl_eqt           1565
pl_orbincl       1522
pl_tranmid       1266
pl_trandep       1777
pl_trandur       1683
pl_rvamp         3667
st_spectype      3847
st_rad           317
st_mass          8
st_met           553
st_lum           311
st_logg          321
st_age           1314
st_dens          586
st_vsin          3881
st_rotp          5268
st_radv          3772
ra               0
dec              0
sy_dist          27
dtype: int64
```

```
In [13]: #we need to drop all planets with missing values under a column, starting with pl_o
required_columns = ['disc_year', 'pl_orbper', 'pl_orbsmax', 'pl_angsep', 'pl_rade',
                    'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl',
                    'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                    'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                    'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data.dropna(subset= required_columns, inplace=True)

#calculating columns with missing values again after dropping planets missing data
#all are zeros
```

```
missing_row_data=new_planet_data.isna()
total_rows=missing_row_data.sum()
print(total_rows)
```

```
pl_name      0
hostname     0
pl_letter    0
sy_snum      0
sy_pnum      0
sy_mnum      0
discoverymethod 0
disc_year    0
disc_facility 0
pl_orbper    0
pl_orbsmax   0
pl_angsep    0
pl_rade      0
pl_bmasse    0
pl_dens      0
pl_orbeccen  0
pl_insol     0
pl_eqt       0
pl_orbincl   0
pl_tranmid   0
pl_trandep   0
pl_trandur   0
pl_rvamp     0
st_spectype  0
st_rad       0
st_mass      0
st_met       0
st_lum       0
st_logg      0
st_age       0
st_dens      0
st_vsin      0
st_rotp      0
st_radv      0
ra           0
dec          0
sy_dist      0
dtype: int64
```

```
In [14]: #calculating rows with missing values again after dropping planets missing data
#all are zeros
missing_column_data = new_planet_data.isna()
total_columns = missing_column_data.sum(axis=1)
#left with 162 entries, or 162 planets that have data in all the required columns
print(total_columns)
```

```
36      0
54      0
55      0
110     0
209     0
..
6078    0
6097    0
6100    0
6107    0
6141    0
```

```
Length: 162, dtype: int64
```

```
In [15]: #check to see if rows are full
         print(new_planet_data)
```

	pl_name	hostname	pl_letter	sy_snum	sy_pnum	sy_mnum	\
36	55 Cnc e	55 Cnc	e	2	7	0	
54	AU Mic b	AU Mic	b	1	4	0	
55	AU Mic c	AU Mic	c	1	4	0	
110	CoRoT-18 b	CoRoT-18	b	1	1	0	
209	G 9-40 b	G 9-40	b	1	1	0	
...	...	...	...	...	...	...	
6078	WASP-96 b	WASP-96	b	1	1	0	
6097	Wolf 327 b	Wolf 327	b	1	1	0	
6100	XO-2 N b	XO-2 N	b	2	3	0	
6107	XO-7 b	XO-7	b	1	1	0	
6141	pi Men c	HD 39091	c	1	3	0	
	discoverymethod	disc_year	\				
36	Radial Velocity	2004					
54	Transit	2020					
55	Transit	2021					
110	Transit	2011					
209	Transit	2019					
...	...	...					
6078	Transit	2014					
6097	Transit	2024					
6100	Transit	2007					
6107	Transit	2019					
6141	Transit	2018					
	disc_facility	pl_orbper	...	st_lum	\		
36	McDonald Observatory	0.736547	...	-0.19695			
54	Transiting Exoplanet Survey Satellite (TESS)	8.463446	...	-0.97757			
55	Transiting Exoplanet Survey Satellite (TESS)	18.859023	...	-0.97757			
110	CoRoT	1.900069	...	-0.17000			
209	K2	5.745998	...	-1.96019			
...	...	...	...	...			
6078	SuperWASP	3.425260	...	0.03226			
6097	Transiting Exoplanet Survey Satellite (TESS)	0.573474	...	-1.62986			
6100	XO	2.615862	...	-0.08823			
6107	XO	2.864142	...	0.43132			
6141	Transiting Exoplanet Survey Satellite (TESS)	6.267840	...	0.15836			
	st_logg	st_age	st_dens	st_vsin	st_rotp	st_radv	ra \
36	4.430	5.500	1.521203	1.23	38.80	27.40949	133.146837
54	4.370	0.019	1.750000	9.20	4.86	-6.31031	311.291137
55	4.370	0.019	1.750000	9.20	4.86	-6.31031	311.291137
110	4.400	0.600	1.350000	8.00	5.40	29.55000	98.172417
209	4.840	9.900	1.620000	2.00	30.00	14.65000	134.718851
...	...	...	...	...	...	...	...
6078	4.420	8.000	1.299000	1.50	20.16	-1.23300	1.046570
6097	4.870	4.100	8.540000	2.00	44.40	9.39000	148.378320
6100	4.433	7.800	1.410000	1.07	28.60	46.93290	117.026769
6107	4.246	1.180	0.451084	6.00	3.62	-12.98300	277.478075
6141	4.430	2.980	0.944584	3.16	18.30	10.70916	84.299280
	dec	sy_dist					
36	28.329815	12.5855					
54	-31.342450	9.7221					
55	-31.342450	9.7221					

```
110    -0.031605  764.8890
209    21.074796   27.9276
...      ...      ...
6078 -47.360635  352.4640
6097  35.570548   28.5321
6100  50.225147  154.2730
6107  85.233321  234.1490
6141 -80.464604   18.2702
```

```
[162 rows x 37 columns]
```

```
In [16]: #redoing the internal index column, 0 to 70
new_planet_data=new_planet_data.reset_index(drop=True)
print(new_planet_data)
```

	pl_name	hostname	pl_letter	sy_snum	sy_pnum	sy_mnum	\
0	55 Cnc e	55 Cnc	e	2	7	0	
1	AU Mic b	AU Mic	b	1	4	0	
2	AU Mic c	AU Mic	c	1	4	0	
3	CoRoT-18 b	CoRoT-18	b	1	1	0	
4	G 9-40 b	G 9-40	b	1	1	0	
..	...	...	...	...	...	...	
157	WASP-96 b	WASP-96	b	1	1	0	
158	Wolf 327 b	Wolf 327	b	1	1	0	
159	XO-2 N b	XO-2 N	b	2	3	0	
160	XO-7 b	XO-7	b	1	1	0	
161	pi Men c	HD 39091	c	1	3	0	

	discoverymethod	disc_year	disc_facility	\
0	Radial Velocity	2004	McDonald Observatory	
1	Transit	2020	Transiting Exoplanet Survey Satellite (TESS)	
2	Transit	2021	Transiting Exoplanet Survey Satellite (TESS)	
3	Transit	2011	CoRoT	
4	Transit	2019	K2	
..	...	...	...	
157	Transit	2014	SuperWASP	
158	Transit	2024	Transiting Exoplanet Survey Satellite (TESS)	
159	Transit	2007	XO	
160	Transit	2019	XO	
161	Transit	2018	Transiting Exoplanet Survey Satellite (TESS)	

	pl_orbper	...	st_lum	st_logg	st_age	st_dens	st_vsin	st_rotp	\
0	0.736547	...	-0.19695	4.430	5.500	1.521203	1.23	38.80	
1	8.463446	...	-0.97757	4.370	0.019	1.750000	9.20	4.86	
2	18.859023	...	-0.97757	4.370	0.019	1.750000	9.20	4.86	
3	1.900069	...	-0.17000	4.400	0.600	1.350000	8.00	5.40	
4	5.745998	...	-1.96019	4.840	9.900	1.620000	2.00	30.00	
..	...	...	...	...	...	...	...	...	
157	3.425260	...	0.03226	4.420	8.000	1.299000	1.50	20.16	
158	0.573474	...	-1.62986	4.870	4.100	8.540000	2.00	44.40	
159	2.615862	...	-0.08823	4.433	7.800	1.410000	1.07	28.60	
160	2.864142	...	0.43132	4.246	1.180	0.451084	6.00	3.62	
161	6.267840	...	0.15836	4.430	2.980	0.944584	3.16	18.30	

	st_radv	ra	dec	sy_dist
0	27.40949	133.146837	28.329815	12.5855
1	-6.31031	311.291137	-31.342450	9.7221
2	-6.31031	311.291137	-31.342450	9.7221
3	29.55000	98.172417	-0.031605	764.8890
4	14.65000	134.718851	21.074796	27.9276
..	...	...	...	...
157	-1.23300	1.046570	-47.360635	352.4640
158	9.39000	148.378320	35.570548	28.5321
159	46.93290	117.026769	50.225147	154.2730
160	-12.98300	277.478075	85.233321	234.1490
161	10.70916	84.299280	-80.464604	18.2702

[162 rows x 37 columns]

```
In [17]: #renaming the planet's id with new table, 1 to length of column, 1 to 71
#new_planet_data['planet_id']=range(1, len(new_planet_data) + 1)
```

```
print(new_planet_data)
```

	pl_name	hostname	pl_letter	sy_snum	sy_pnum	sy_mnum	\
0	55 Cnc e	55 Cnc	e	2	7	0	
1	AU Mic b	AU Mic	b	1	4	0	
2	AU Mic c	AU Mic	c	1	4	0	
3	CoRoT-18 b	CoRoT-18	b	1	1	0	
4	G 9-40 b	G 9-40	b	1	1	0	
..	...	...	...	...	...	...	
157	WASP-96 b	WASP-96	b	1	1	0	
158	Wolf 327 b	Wolf 327	b	1	1	0	
159	XO-2 N b	XO-2 N	b	2	3	0	
160	XO-7 b	XO-7	b	1	1	0	
161	pi Men c	HD 39091	c	1	3	0	

	discoverymethod	disc_year	disc_facility	\
0	Radial Velocity	2004	McDonald Observatory	
1	Transit	2020	Transiting Exoplanet Survey Satellite (TESS)	
2	Transit	2021	Transiting Exoplanet Survey Satellite (TESS)	
3	Transit	2011	CoRoT	
4	Transit	2019	K2	
..	...	...	...	
157	Transit	2014	SuperWASP	
158	Transit	2024	Transiting Exoplanet Survey Satellite (TESS)	
159	Transit	2007	XO	
160	Transit	2019	XO	
161	Transit	2018	Transiting Exoplanet Survey Satellite (TESS)	

	pl_orbper	...	st_lum	st_logg	st_age	st_dens	st_vsin	st_rotp	\
0	0.736547	...	-0.19695	4.430	5.500	1.521203	1.23	38.80	
1	8.463446	...	-0.97757	4.370	0.019	1.750000	9.20	4.86	
2	18.859023	...	-0.97757	4.370	0.019	1.750000	9.20	4.86	
3	1.900069	...	-0.17000	4.400	0.600	1.350000	8.00	5.40	
4	5.745998	...	-1.96019	4.840	9.900	1.620000	2.00	30.00	
..	...	...	...	...	...	...	...	...	
157	3.425260	...	0.03226	4.420	8.000	1.299000	1.50	20.16	
158	0.573474	...	-1.62986	4.870	4.100	8.540000	2.00	44.40	
159	2.615862	...	-0.08823	4.433	7.800	1.410000	1.07	28.60	
160	2.864142	...	0.43132	4.246	1.180	0.451084	6.00	3.62	
161	6.267840	...	0.15836	4.430	2.980	0.944584	3.16	18.30	

	st_radv	ra	dec	sy_dist
0	27.40949	133.146837	28.329815	12.5855
1	-6.31031	311.291137	-31.342450	9.7221
2	-6.31031	311.291137	-31.342450	9.7221
3	29.55000	98.172417	-0.031605	764.8890
4	14.65000	134.718851	21.074796	27.9276
..	...	...	...	...
157	-1.23300	1.046570	-47.360635	352.4640
158	9.39000	148.378320	35.570548	28.5321
159	46.93290	117.026769	50.225147	154.2730
160	-12.98300	277.478075	85.233321	234.1490
161	10.70916	84.299280	-80.464604	18.2702

```
[162 rows x 37 columns]
```

Stage 2. Model your data

```
In [18]: ## instead of having two datasets, make the columns names into their full names for  
  
#rename columns here  
new_planet_data = new_planet_data.rename(columns={'pl_name': 'Planet Name',  
                                                'hostname': 'Host Name', 'pl_lett  
                                                'sy_snum': 'Number of Stars', 'sy_  
                                                'sy_mnum': 'Number of Moons', 'disc  
                                                'disc_year': 'Discovery Year', 'dis  
                                                'pl_orbper': 'Orbital Period [days  
                                                'pl_angsep': 'Angular Separation [  
                                                'pl_bmasse': 'Planet Mass or Mass*  
                                                'pl_orbeccen': 'Eccentricity', 'p  
                                                'pl_eqt': 'Equilibrium Temperatur  
                                                'pl_tranmid': 'Transit Midpoint [d  
                                                'pl_trandur': 'Transit Duration [h  
                                                'st_spectype': 'Spectral Type', 's  
                                                'st_mass': 'Stellar Mass [Solar ma  
                                                'st_lum': 'Stellar Luminosity [log  
                                                'st_age': 'Stellar Age [Gyr]', 'st  
                                                'st_vsin': 'Stellar Rotational Vel  
                                                'st_radv': 'Systemic Radial Veloci  
                                                'sy_dist': 'Distance [pc]'})  
  
print(new_planet_data)
```

	Planet Name	Host Name	Planet Letter	Number of Stars	Number of Planets	\
0	55 Cnc e	55 Cnc	e	2	7	
1	AU Mic b	AU Mic	b	1	4	
2	AU Mic c	AU Mic	c	1	4	
3	CoRoT-18 b	CoRoT-18	b	1	1	
4	G 9-40 b	G 9-40	b	1	1	
..	...	...	...	...	...	
157	WASP-96 b	WASP-96	b	1	1	
158	Wolf 327 b	Wolf 327	b	1	1	
159	XO-2 N b	XO-2 N	b	2	3	
160	XO-7 b	XO-7	b	1	1	
161	pi Men c	HD 39091	c	1	3	

	Number of Moons	Discovery Method	Discovery Year	\
0	0	Radial Velocity	2004	
1	0	Transit	2020	
2	0	Transit	2021	
3	0	Transit	2011	
4	0	Transit	2019	
..	...	...	...	
157	0	Transit	2014	
158	0	Transit	2024	
159	0	Transit	2007	
160	0	Transit	2019	
161	0	Transit	2018	

	Discovery Facility	Orbital Period [days]	...	\
0	McDonald Observatory	0.736547	...	
1	Transiting Exoplanet Survey Satellite (TESS)	8.463446	...	
2	Transiting Exoplanet Survey Satellite (TESS)	18.859023	...	
3	CoRoT	1.900069	...	
4	K2	5.745998	...	
..	...	...	...	
157	SuperWASP	3.425260	...	
158	Transiting Exoplanet Survey Satellite (TESS)	0.573474	...	
159	XO	2.615862	...	
160	XO	2.864142	...	
161	Transiting Exoplanet Survey Satellite (TESS)	6.267840	...	

	Stellar Luminosity [log(Solar)]	\
0	-0.19695	
1	-0.97757	
2	-0.97757	
3	-0.17000	
4	-1.96019	
..	...	
157	0.03226	
158	-1.62986	
159	-0.08823	
160	0.43132	
161	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
2	4.370	0.019	

3	4.400	0.600
4	4.840	9.900
..	...	...
157	4.420	8.000
158	4.870	4.100
159	4.433	7.800
160	4.246	1.180
161	4.430	2.980

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s] \
0	1.521203	1.23
1	1.750000	9.20
2	1.750000	9.20
3	1.350000	8.00
4	1.620000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s] \
0	38.80	27.40949
1	4.86	-6.31031
2	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

	RA [deg]	Dec [deg]	Distance [pc]
0	133.146837	28.329815	12.5855
1	311.291137	-31.342450	9.7221
2	311.291137	-31.342450	9.7221
3	98.172417	-0.031605	764.8890
4	134.718851	21.074796	27.9276
..	...	...	...
157	1.046570	-47.360635	352.4640
158	148.378320	35.570548	28.5321
159	117.026769	50.225147	154.2730
160	277.478075	85.233321	234.1490
161	84.299280	-80.464604	18.2702

[162 rows x 37 columns]

```
In [19]: ## ***** foreign keys columns do not have 1 to n in sequential order *****

##add new columns to dataset, Star ID, Discovery ID, Solar System ID

#variables for number of columns and rows or current data
```

```
#number_columns = len(new_planet_data.columns)
#number_rows = len(new_planet_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#new_planet_data.insert(number_columns, 'Star ID', range(1, number_rows + 1))

#new_planet_data.insert(number_columns + 1, 'System ID', range(1, number_rows + 1))

#new_planet_data.insert(number_columns + 2, 'Discovery ID', range(1, number_rows +

print(new_planet_data)
```

	Planet Name	Host Name	Planet Letter	Number of Stars	Number of Planets	\
0	55 Cnc e	55 Cnc	e	2	7	
1	AU Mic b	AU Mic	b	1	4	
2	AU Mic c	AU Mic	c	1	4	
3	CoRoT-18 b	CoRoT-18	b	1	1	
4	G 9-40 b	G 9-40	b	1	1	
..	...	...	...	...	...	
157	WASP-96 b	WASP-96	b	1	1	
158	Wolf 327 b	Wolf 327	b	1	1	
159	XO-2 N b	XO-2 N	b	2	3	
160	XO-7 b	XO-7	b	1	1	
161	pi Men c	HD 39091	c	1	3	

	Number of Moons	Discovery Method	Discovery Year	\
0	0	Radial Velocity	2004	
1	0	Transit	2020	
2	0	Transit	2021	
3	0	Transit	2011	
4	0	Transit	2019	
..	...	...	...	
157	0	Transit	2014	
158	0	Transit	2024	
159	0	Transit	2007	
160	0	Transit	2019	
161	0	Transit	2018	

	Discovery Facility	Orbital Period [days]	...	\
0	McDonald Observatory	0.736547	...	
1	Transiting Exoplanet Survey Satellite (TESS)	8.463446	...	
2	Transiting Exoplanet Survey Satellite (TESS)	18.859023	...	
3	CoRoT	1.900069	...	
4	K2	5.745998	...	
..	...	...	...	
157	SuperWASP	3.425260	...	
158	Transiting Exoplanet Survey Satellite (TESS)	0.573474	...	
159	XO	2.615862	...	
160	XO	2.864142	...	
161	Transiting Exoplanet Survey Satellite (TESS)	6.267840	...	

	Stellar Luminosity [log(Solar)]	\
0	-0.19695	
1	-0.97757	
2	-0.97757	
3	-0.17000	
4	-1.96019	
..	...	
157	0.03226	
158	-1.62986	
159	-0.08823	
160	0.43132	
161	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
2	4.370	0.019	

3	4.400	0.600
4	4.840	9.900
..	...	...
157	4.420	8.000
158	4.870	4.100
159	4.433	7.800
160	4.246	1.180
161	4.430	2.980

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s] \
0	1.521203	1.23
1	1.750000	9.20
2	1.750000	9.20
3	1.350000	8.00
4	1.620000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s] \
0	38.80	27.40949
1	4.86	-6.31031
2	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

	RA [deg]	Dec [deg]	Distance [pc]
0	133.146837	28.329815	12.5855
1	311.291137	-31.342450	9.7221
2	311.291137	-31.342450	9.7221
3	98.172417	-0.031605	764.8890
4	134.718851	21.074796	27.9276
..	...	...	...
157	1.046570	-47.360635	352.4640
158	148.378320	35.570548	28.5321
159	117.026769	50.225147	154.2730
160	277.478075	85.233321	234.1490
161	84.299280	-80.464604	18.2702

[162 rows x 37 columns]

```
In [20]: ## Create a new dataset for Stars, Solar System, and Discoveries

#stars_data = new_planet_data['Star ID']
#print(stars_data)
```

```
#system_data = new_planet_data['System ID']
#print(system_data)

#discoveries_data = new_planet_data['Discovery ID']
#print(discoveries_data)
```

```
In [21]: ## Add data to new data sets
#i need to find the columns, then i need add the columns to the new dataset

#Stars
#Spectral Type, Stellar Radius [Solar Radius],
#Stellar Mass [Solar mass], Stellar Metallicity [dex],
#Stellar Luminosity [log(Solar)], Stellar Surface Gravity [log10(cm/s**2)],
#Stellar Age [Gyr], Stellar Density [g/cm**3], Stellar Rotational Velocity [km/s]
#Stellar Rotational Period [days], Systemic Radial Velocity [km/s],

#find columns and assign to variable
stars_columns=['Host Name','Spectral Type', 'Stellar Radius [Solar Radius]',
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]']

#redefine stars_data using new variable with column names
stars_data = new_planet_data[stars_columns]

#set variable number of rows in dataset
#number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#stars_data.insert(0, 'Star ID', range(1, number_rows + 1))

print(stars_data)
```

	Host Name	Spectral Type	Stellar Radius [Solar Radius]	\
0	55 Cnc	G8 V	0.9430	
1	AU Mic	M1 V	0.8620	
2	AU Mic	M1 V	0.8620	
3	CoRoT-18	G9 V	1.0000	
4	G 9-40	M2.5 V	0.3026	
..	...	...	...	
157	WASP-96	G8	1.0500	
158	Wolf 327	M2.5 V	0.4060	
159	XO-2 N	K0 V	0.9900	
160	XO-7	G0 V	1.4800	
161	HD 39091	G0 V	1.1700	

	Stellar Mass [Solar mass]	Stellar Metallicity [dex]	\
0	0.9050	0.350	
1	0.6350	0.010	
2	0.6350	0.010	
3	0.9500	-0.100	
4	0.2952	-0.070	
..	...	...	
157	1.0600	0.140	
158	0.4050	-0.170	
159	0.9710	0.430	
160	1.4050	0.432	
161	1.0700	0.090	

	Stellar Luminosity [log(Solar)]	\
0	-0.19695	
1	-0.97757	
2	-0.97757	
3	-0.17000	
4	-1.96019	
..	...	
157	0.03226	
158	-1.62986	
159	-0.08823	
160	0.43132	
161	0.15836	

	Stellar Surface Gravity [log ₁₀ (cm/s ²)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
2	4.370	0.019	
3	4.400	0.600	
4	4.840	9.900	
..	...	...	
157	4.420	8.000	
158	4.870	4.100	
159	4.433	7.800	
160	4.246	1.180	
161	4.430	2.980	

	Stellar Density [g/cm ³]	Stellar Rotational Velocity [km/s]	\
0	1.521203	1.23	
1	1.750000	9.20	
2	1.750000	9.20	

3	1.350000	8.00
4	1.620000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s]
0	38.80	27.40949
1	4.86	-6.31031
2	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

[162 rows x 12 columns]

```
In [22]: ## there is a foreign key in stars data that links it to solar system
# we need to add system_id at the end of the dataset like we need for exoplanets ab

number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows)
stars_data.insert(1, 'System ID', range(1, number_rows + 1))

print(stars_data)
```

	Host Name	System ID	Spectral Type	Stellar Radius [Solar Radius]	\
0	55 Cnc	1	G8 V	0.9430	
1	AU Mic	2	M1 V	0.8620	
2	AU Mic	3	M1 V	0.8620	
3	CoRoT-18	4	G9 V	1.0000	
4	G 9-40	5	M2.5 V	0.3026	
..	...	...	...	...	
157	WASP-96	158	G8	1.0500	
158	Wolf 327	159	M2.5 V	0.4060	
159	XO-2 N	160	K0 V	0.9900	
160	XO-7	161	G0 V	1.4800	
161	HD 39091	162	G0 V	1.1700	

	Stellar Mass [Solar mass]	Stellar Metallicity [dex]	\
0	0.9050	0.350	
1	0.6350	0.010	
2	0.6350	0.010	
3	0.9500	-0.100	
4	0.2952	-0.070	
..	...	...	
157	1.0600	0.140	
158	0.4050	-0.170	
159	0.9710	0.430	
160	1.4050	0.432	
161	1.0700	0.090	

	Stellar Luminosity [log(Solar)]	\
0	-0.19695	
1	-0.97757	
2	-0.97757	
3	-0.17000	
4	-1.96019	
..	...	
157	0.03226	
158	-1.62986	
159	-0.08823	
160	0.43132	
161	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
2	4.370	0.019	
3	4.400	0.600	
4	4.840	9.900	
..	...	...	
157	4.420	8.000	
158	4.870	4.100	
159	4.433	7.800	
160	4.246	1.180	
161	4.430	2.980	

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s]	\
0	1.521203	1.23	
1	1.750000	9.20	
2	1.750000	9.20	

3	1.350000	8.00
4	1.620000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s]
0	38.80	27.40949
1	4.86	-6.31031
2	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

[162 rows x 13 columns]

```
In [23]: #System
#System ID, Host Name, Number of Stars, Number of Planets, RA [deg],
#Dec [deg], Distance [pc]

system_columns=[ 'Host Name', 'Number of Stars', 'Number of Planets', 'RA [deg]',
'Dec [deg]', 'Distance [pc]']

system_data = new_planet_data[system_columns]

#set variable number of rows in dataset
#number_rows = len(system_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#system_data.insert(0, 'System ID', range(1, number_rows + 1))

print(system_data)
```

	Host Name	Number of Stars	Number of Planets	RA [deg]	Dec [deg]	\
0	55 Cnc	2	7	133.146837	28.329815	
1	AU Mic	1	4	311.291137	-31.342450	
2	AU Mic	1	4	311.291137	-31.342450	
3	CoRoT-18	1	1	98.172417	-0.031605	
4	G 9-40	1	1	134.718851	21.074796	
..	...	...	...	...	...	
157	WASP-96	1	1	1.046570	-47.360635	
158	Wolf 327	1	1	148.378320	35.570548	
159	XO-2 N	2	3	117.026769	50.225147	
160	XO-7	1	1	277.478075	85.233321	
161	HD 39091	1	3	84.299280	-80.464604	

	Distance [pc]
0	12.5855
1	9.7221
2	9.7221
3	764.8890
4	27.9276
..	...
157	352.4640
158	28.5321
159	154.2730
160	234.1490
161	18.2702

[162 rows x 6 columns]

```
In [24]: #Discoveries
#Discovery ID, Discovery Method, Discovery Year, Discovery Facility

discovery_columns=['Host Name','Discovery Facility','Discovery Year','Discovery Met

discoveries_data = new_planet_data[discovery_columns]

#set variable number of rows in dataset
#number_rows = len(discoveries_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#discoveries_data.insert(0, 'Discovery ID', range(1, number_rows + 1))

print(discoveries_data)
```

	Host Name	Discovery Facility	Discovery Year	\
0	55 Cnc	McDonald Observatory	2004	
1	AU Mic	Transiting Exoplanet Survey Satellite (TESS)	2020	
2	AU Mic	Transiting Exoplanet Survey Satellite (TESS)	2021	
3	CoRoT-18	CoRoT	2011	
4	G 9-40	K2	2019	
..	...	...	...	
157	WASP-96	SuperWASP	2014	
158	Wolf 327	Transiting Exoplanet Survey Satellite (TESS)	2024	
159	XO-2 N	XO	2007	
160	XO-7	XO	2019	
161	HD 39091	Transiting Exoplanet Survey Satellite (TESS)	2018	

	Discovery Method
0	Radial Velocity
1	Transit
2	Transit
3	Transit
4	Transit
..	...
157	Transit
158	Transit
159	Transit
160	Transit
161	Transit

[162 rows x 4 columns]

```
In [25]: #Planet ID, Planet Name, Planet Letter, Number of Moons, Orbital Period [days], Orb
#Angular Separation [mas], Planet Radius [Earth Radius], Planet Mass or Mass*sin(i)
#Eccentricity, Insolation Flux [Earth Flux], Equilibrium Temperature [K], Inclinati
#Transit Depth [%], Transit Duration [hours], Radial Velocity Amplitude [m/s], Star

planet_columns=['Host Name','Planet Name', 'Planet Letter', 'Number of Moons', 'Orb
'Angular Separation [mas]', 'Planet Radius [Earth Radius]', 'Planet Mass or Mass*si
'Eccentricity', 'Insolation Flux [Earth Flux]', 'Equilibrium Temperature [K]', 'Inc
'Transit Depth [%]', 'Transit Duration [hours]', 'Radial Velocity Amplitude [m/s]']

new_planet_data = new_planet_data[planet_columns]
print(new_planet_data)
```

	Host Name	Planet Name	Planet Letter	Number of Moons	\
0	55 Cnc	55 Cnc e	e	0	
1	AU Mic	AU Mic b	b	0	
2	AU Mic	AU Mic c	c	0	
3	CoRoT-18	CoRoT-18 b	b	0	
4	G 9-40	G 9-40 b	b	0	
..	...	...	...	...	
157	WASP-96	WASP-96 b	b	0	
158	Wolf 327	Wolf 327 b	b	0	
159	XO-2 N	XO-2 N b	b	0	
160	XO-7	XO-7 b	b	0	
161	HD 39091	pi Men c	c	0	

	Orbital Period [days]	Orbit Semi-Major Axis [au]	\
0	0.736547	0.01544	
1	8.463446	0.07000	
2	18.859023	0.11900	
3	1.900069	0.02950	
4	5.745998	0.04180	
..	...	...	
157	3.425260	0.04530	
158	0.573474	0.01000	
159	2.615862	0.03680	
160	2.864142	0.04421	
161	6.267840	0.06839	

	Angular Separation [mas]	Planet Radius [Earth Radius]	\
0	1.2300	1.875000	
1	7.2000	4.790000	
2	12.2000	2.790000	
3	0.0386	14.680000	
4	1.5000	1.900000	
..	...	...	
157	0.1290	13.450000	
158	0.3500	1.240000	
159	0.2390	11.130537	
160	0.1890	15.389957	
161	3.7400	2.018900	

	Planet Mass or Mass*sin(i) [Earth Mass]	Planet Density [g/cm**3]	\
0	7.99000	6.660	
1	8.99000	0.490	
2	14.46000	3.900	
3	1102.82000	2.200	
4	4.00000	3.200	
..	...	...	
157	152.55000	0.370	
158	2.53000	7.240	
159	179.89178	0.715	
160	225.34147	0.340	
161	3.63000	2.800	

	Eccentricity	Insolation Flux [Earth Flux]	Equilibrium Temperature [K]	\
0	0.050	2657.8322	1958.0	
1	0.070	21.2000	554.8	
2	0.180	7.3000	424.7	

3	0.080	823.7009	1550.0
4	0.000	6.2700	440.6
..	...	...	...
157	0.000	517.0000	1285.0
158	0.000	233.9000	996.0
159	0.028	428.0360	1328.0
160	0.038	1540.0000	1743.0
161	0.000	309.0000	1169.8

	Inclination [deg]	Transit Midpoint [days]	Transit Depth [%] \
0	83.590	2457063.210	0.038467
1	88.390	2458330.351	0.249000
2	89.460	2458342.223	0.085000
3	86.500	2455321.724	2.571701
4	89.030	2459503.327	0.365000
..	...	...	...
157	85.600	2456258.062	1.380000
158	84.890	2459252.981	0.081839
159	88.010	2454508.738	1.336300
160	83.450	2457917.475	0.909000
161	87.045	2458425.789	0.026759

	Transit Duration [hours]	Radial Velocity Amplitude [m/s]
0	1.543900	6.020
1	3.240000	3.800
2	4.290000	4.696
3	2.387000	590.000
4	1.693000	3.220
..	...	...
157	2.426400	62.000
158	0.900451	3.540
159	2.683900	90.170
160	2.772000	80.500
161	2.952024	1.185

[162 rows x 18 columns]

```
In [26]: #check for missing values or Na in each column,
#True for missing, false for NOT missing

missing_values = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('_____Exoplanets_____')

missing_values = stars_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('_____Stars_____')

missing_values = system_data.isna()
```

```
row_total=missing_values.sum()
print(row_total)
print('_____Solar System_____')

missing_values = discoveries_data.isna()

row_total=missing_values.sum()
print(row_total)
print('_____Discoveries_____')
```

Host Name	0
Planet Name	0
Planet Letter	0
Number of Moons	0
Orbital Period [days]	0
Orbit Semi-Major Axis [au]	0
Angular Separation [mas]	0
Planet Radius [Earth Radius]	0
Planet Mass or Mass*sin(i) [Earth Mass]	0
Planet Density [g/cm**3]	0
Eccentricity	0
Insolation Flux [Earth Flux]	0
Equilibrium Temperature [K]	0
Inclination [deg]	0
Transit Midpoint [days]	0
Transit Depth [%]	0
Transit Duration [hours]	0
Radial Velocity Amplitude [m/s]	0

dtype: int64

Exoplanets

Host Name	0
System ID	0
Spectral Type	0
Stellar Radius [Solar Radius]	0
Stellar Mass [Solar mass]	0
Stellar Metallicity [dex]	0
Stellar Luminosity [log(Solar)]	0
Stellar Surface Gravity [log10(cm/s**2)]	0
Stellar Age [Gyr]	0
Stellar Density [g/cm**3]	0
Stellar Rotational Velocity [km/s]	0
Stellar Rotational Period [days]	0
Systemic Radial Velocity [km/s]	0

dtype: int64

Stars

Host Name	0
Number of Stars	0
Number of Planets	0
RA [deg]	0
Dec [deg]	0
Distance [pc]	0

dtype: int64

Solar System

Host Name	0
Discovery Facility	0
Discovery Year	0
Discovery Method	0

dtype: int64

Discoveries

```
In [27]: #count of entries before drop_duplicates
star_entries_before = len(stars_data)
print( 'stars before: ', star_entries_before)

stars_data.drop_duplicates(subset=['Spectral Type', 'Stellar Radius [Solar Radius]',
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
```

```
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]'], inplace=True

#after drops
star_entries_after = len(stars_data)
print( 'stars after:', star_entries_after)

#difference
star_entries_dropped = star_entries_before - star_entries_after
print("Duplicates Removed:", star_entries_dropped)
```

stars before: 162

stars after: 131

Duplicates Removed: 31

```
In [28]: #set variable number of rows in dataset
number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows) at the end of the
stars_data.insert(0, 'Star ID', range(1, number_rows + 1))

print(stars_data)
```

	Star ID	Host Name	System ID	Spectral Type	\
0	1	55 Cnc	1	G8 V	
1	2	AU Mic	2	M1 V	
3	3	CoRoT-18	4	G9 V	
4	4	G 9-40	5	M2.5 V	
5	5	GJ 1132	6	M4.5 V	
..	...	...	...	...	
157	127	WASP-96	158	G8	
158	128	Wolf 327	159	M2.5 V	
159	129	XO-2 N	160	K0 V	
160	130	XO-7	161	G0 V	
161	131	HD 39091	162	G0 V	

	Stellar Radius [Solar Radius]	Stellar Mass [Solar mass]	\
0	0.9430	0.9050	
1	0.8620	0.6350	
3	1.0000	0.9500	
4	0.3026	0.2952	
5	0.2211	0.1945	
..	...	...	
157	1.0500	1.0600	
158	0.4060	0.4050	
159	0.9900	0.9710	
160	1.4800	1.4050	
161	1.1700	1.0700	

	Stellar Metallicity [dex]	Stellar Luminosity [log(Solar)]	\
0	0.350	-0.19695	
1	0.010	-0.97757	
3	-0.100	-0.17000	
4	-0.070	-1.96019	
5	-0.170	-2.32148	
..	...	...	
157	0.140	0.03226	
158	-0.170	-1.62986	
159	0.430	-0.08823	
160	0.432	0.43132	
161	0.090	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
3	4.400	0.600	
4	4.840	9.900	
5	5.037	5.000	
..	...	...	
157	4.420	8.000	
158	4.870	4.100	
159	4.433	7.800	
160	4.246	1.180	
161	4.430	2.980	

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s]	\
0	1.521203	1.23	
1	1.750000	9.20	
3	1.350000	8.00	

4	1.620000	2.00
5	25.300000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s]
0	38.80	27.40949
1	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
5	122.30	35.07880
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

[131 rows x 14 columns]

```
In [29]: #count of entries before drop_duplicates
system_entries_before = len(system_data)
print('system before: ', system_entries_before)

system_data.drop_duplicates(subset=['Host Name'], inplace=True)

#after
system_entries_after = len(system_data)
print('system after:', system_entries_after)

#difference
system_entries_dropped = system_entries_before - system_entries_after

print("Duplicates Removed:", system_entries_dropped)
```

system before: 162
system after: 130
Duplicates Removed: 32

```
In [30]: #set variable number of rows in dataset
number_rows = len(system_data)

## add new columns using data.insert(location, column name, rows) at the end of the
system_data.insert(0, 'System ID', range(1, number_rows + 1))

print(system_data)
```

	System ID	Host Name	Number of Stars	Number of Planets	RA [deg]	\
0	1	55 Cnc	2	7	133.146837	
1	2	AU Mic	1	4	311.291137	
3	3	CoRoT-18	1	1	98.172417	
4	4	G 9-40	1	1	134.718851	
5	5	GJ 1132	1	2	153.709070	
..	...	...	...	...	...	
157	126	WASP-96	1	1	1.046570	
158	127	Wolf 327	1	1	148.378320	
159	128	XO-2 N	2	3	117.026769	
160	129	XO-7	1	1	277.478075	
161	130	HD 39091	1	3	84.299280	

	Dec [deg]	Distance [pc]
0	28.329815	12.5855
1	-31.342450	9.7221
3	-0.031605	764.8890
4	21.074796	27.9276
5	-47.154936	12.6130
..	...	...
157	-47.360635	352.4640
158	35.570548	28.5321
159	50.225147	154.2730
160	85.233321	234.1490
161	-80.464604	18.2702

[130 rows x 7 columns]

```
In [31]: #count of entries before drop_duplicates
discovery_entries_before = len(discoveries_data)
print('discoveries before: ', discovery_entries_before)

discoveries_data.drop_duplicates(subset=[ 'Discovery Facility','Discovery Year','Di

#after
discovery_entries_after = len(discoveries_data)
print( 'discoveries after:', discovery_entries_after)

#difference
discovery_entries_dropped = discovery_entries_before - discovery_entries_after

print("Duplicates Removed:", discovery_entries_dropped)
```

```
discoveries before: 162
discoveries after: 56
Duplicates Removed: 106
```

```
In [32]: #set variable number of rows in dataset
number_rows = len(discoveries_data)

## add new columns using data.insert(location, column name, rows) at the end of the
discoveries_data.insert(0, 'Discovery ID', range(1, number_rows + 1))

print(discoveries_data)
```

	Discovery ID	Host Name	Discovery Facility \
0	1	55 Cnc	McDonald Observatory
1	2	AU Mic	Transiting Exoplanet Survey Satellite (TESS)
2	3	AU Mic	Transiting Exoplanet Survey Satellite (TESS)
3	4	CoRoT-18	CoRoT
4	5	G 9-40	K2
5	6	GJ 1132	Cerro Tololo Inter-American Observatory
6	7	GJ 1214	MEarth Project
7	8	GJ 3929	Transiting Exoplanet Survey Satellite (TESS)
8	9	GJ 436	W. M. Keck Observatory
9	10	GJ 486	Calar Alto Observatory
10	11	GJ 9827	K2
13	12	Gliese 12	Transiting Exoplanet Survey Satellite (TESS)
14	13	HAT-P-11	HATNet
15	14	HAT-P-19	HATNet
19	15	HAT-P-3	HATNet
20	16	HATS-16	HATSouth
21	17	HATS-18	Multiple Observatories
24	18	HD 118203	Haute-Provence Observatory
25	19	HD 136352	La Silla Observatory
28	20	HD 15337	Transiting Exoplanet Survey Satellite (TESS)
30	21	HD 17156	Subaru Telescope
34	22	HD 209458	W. M. Keck Observatory
38	23	HD 3167	K2
42	24	HD 73344	K2
56	25	K2-141	K2
58	26	K2-19	K2
66	27	KELT-18	KELT-North
67	28	Kepler-103	Kepler
69	29	Kepler-17	Kepler
70	30	Kepler-21	Kepler
74	31	Kepler-539	Kepler
75	32	Kepler-63	Kepler
85	33	Qatar-3	Qatar
89	34	Qatar-7	Qatar
101	35	TOI-1416	Transiting Exoplanet Survey Satellite (TESS)
105	36	TOI-2093	Transiting Exoplanet Survey Satellite (TESS)
127	37	WASP-10	SuperWASP
128	38	WASP-107	SuperWASP-South
129	39	WASP-113	SuperWASP-North
130	40	WASP-118	SuperWASP
132	41	WASP-124	SuperWASP-South
135	42	WASP-148	SuperWASP-North
136	43	WASP-166	WASP-South
137	44	WASP-172	SuperWASP-South
139	45	WASP-180 A	SuperWASP
141	46	WASP-186	SuperWASP
142	47	WASP-19	SuperWASP
143	48	WASP-22	SuperWASP
144	49	WASP-4	SuperWASP
145	50	WASP-43	SuperWASP
147	51	WASP-47	SuperWASP
151	52	WASP-69	SuperWASP
154	53	WASP-89	SuperWASP
159	54	XO-2 N	XO
160	55	XO-7	XO

161 56 HD 39091 Transiting Exoplanet Survey Satellite (TESS)

	Discovery Year	Discovery Method
0	2004	Radial Velocity
1	2020	Transit
2	2021	Transit
3	2011	Transit
4	2019	Transit
5	2015	Transit
6	2009	Transit
7	2022	Transit
8	2004	Radial Velocity
9	2021	Radial Velocity
10	2017	Transit
13	2024	Transit
14	2008	Transit
15	2010	Transit
19	2007	Transit
20	2016	Transit
21	2016	Transit
24	2005	Radial Velocity
25	2019	Radial Velocity
28	2019	Transit
30	2007	Radial Velocity
34	1999	Radial Velocity
38	2016	Transit
42	2024	Transit
56	2018	Transit
58	2015	Transit
66	2017	Transit
67	2014	Transit
69	2011	Transit
70	2012	Transit
74	2016	Transit
75	2013	Transit
85	2017	Transit
89	2019	Transit
101	2023	Transit
105	2025	Transit
127	2008	Transit
128	2017	Transit
129	2016	Transit
130	2016	Transit
132	2016	Transit
135	2020	Transit
136	2019	Transit
137	2018	Transit
139	2019	Transit
141	2020	Transit
142	2009	Transit
143	2010	Transit
144	2007	Transit
145	2011	Transit
147	2012	Transit
151	2014	Transit
154	2015	Transit

159	2007	Transit
160	2019	Transit
161	2018	Transit

```
In [33]: #count of entries before drop_duplicates
planet_entries_before = len(new_planet_data)
print('planets before: ', planet_entries_before)

new_planet_data.drop_duplicates(subset=['Planet Name'], inplace=True)

#after
planet_entries_after = len(new_planet_data)
print('planet after:', planet_entries_after)

#difference
planet_entries_dropped = planet_entries_before - planet_entries_after

print("Duplicates Removed:", planet_entries_dropped)
```

planets before: 162
planet after: 162
Duplicates Removed: 0

```
In [34]: #set variable number of rows in dataset
number_rows = len(new_planet_data)

new_planet_data.insert(0, 'Planet ID', range(1, number_rows+1))

print(new_planet_data)
```

	Planet ID	Host Name	Planet Name	Planet Letter	Number of Moons	\
0	1	55 Cnc	55 Cnc e	e	0	
1	2	AU Mic	AU Mic b	b	0	
2	3	AU Mic	AU Mic c	c	0	
3	4	CoRoT-18	CoRoT-18 b	b	0	
4	5	G 9-40	G 9-40 b	b	0	
..	...	...	...	...	...	
157	158	WASP-96	WASP-96 b	b	0	
158	159	Wolf 327	Wolf 327 b	b	0	
159	160	XO-2 N	XO-2 N b	b	0	
160	161	XO-7	XO-7 b	b	0	
161	162	HD 39091	pi Men c	c	0	

	Orbital Period [days]	Orbit Semi-Major Axis [au]	\
0	0.736547	0.01544	
1	8.463446	0.07000	
2	18.859023	0.11900	
3	1.900069	0.02950	
4	5.745998	0.04180	
..	...	...	
157	3.425260	0.04530	
158	0.573474	0.01000	
159	2.615862	0.03680	
160	2.864142	0.04421	
161	6.267840	0.06839	

	Angular Separation [mas]	Planet Radius [Earth Radius]	\
0	1.2300	1.875000	
1	7.2000	4.790000	
2	12.2000	2.790000	
3	0.0386	14.680000	
4	1.5000	1.900000	
..	...	...	
157	0.1290	13.450000	
158	0.3500	1.240000	
159	0.2390	11.130537	
160	0.1890	15.389957	
161	3.7400	2.018900	

	Planet Mass or Mass*sin(i) [Earth Mass]	Planet Density [g/cm**3]	\
0	7.99000	6.660	
1	8.99000	0.490	
2	14.46000	3.900	
3	1102.82000	2.200	
4	4.00000	3.200	
..	...	...	
157	152.55000	0.370	
158	2.53000	7.240	
159	179.89178	0.715	
160	225.34147	0.340	
161	3.63000	2.800	

	Eccentricity	Insolation Flux [Earth Flux]	Equilibrium Temperature [K]	\
0	0.050	2657.8322	1958.0	
1	0.070	21.2000	554.8	
2	0.180	7.3000	424.7	

3	0.080	823.7009	1550.0
4	0.000	6.2700	440.6
..	...	...	...
157	0.000	517.0000	1285.0
158	0.000	233.9000	996.0
159	0.028	428.0360	1328.0
160	0.038	1540.0000	1743.0
161	0.000	309.0000	1169.8

	Inclination [deg]	Transit Midpoint [days]	Transit Depth [%]	\
0	83.590	2457063.210	0.038467	
1	88.390	2458330.351	0.249000	
2	89.460	2458342.223	0.085000	
3	86.500	2455321.724	2.571701	
4	89.030	2459503.327	0.365000	
..	...	...	...	
157	85.600	2456258.062	1.380000	
158	84.890	2459252.981	0.081839	
159	88.010	2454508.738	1.336300	
160	83.450	2457917.475	0.909000	
161	87.045	2458425.789	0.026759	

	Transit Duration [hours]	Radial Velocity Amplitude [m/s]
0	1.543900	6.020
1	3.240000	3.800
2	4.290000	4.696
3	2.387000	590.000
4	1.693000	3.220
..	...	...
157	2.426400	62.000
158	0.900451	3.540
159	2.683900	90.170
160	2.772000	80.500
161	2.952024	1.185

[162 rows x 19 columns]

```
In [35]: #check for missing values or Na in each column,
#True for missing, false for NOT missing

missing_values = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('_____Exoplanets_____')

missing_values = stars_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('_____Stars_____')

missing_values = system_data.isna()
```

```
row_total=missing_values.sum()
print(row_total)
print('_____Solar System_____')

missing_values = discoveries_data.isna()

row_total=missing_values.sum()
print(row_total)
print('_____Discoveries_____')
```

Planet ID	0
Host Name	0
Planet Name	0
Planet Letter	0
Number of Moons	0
Orbital Period [days]	0
Orbit Semi-Major Axis [au]	0
Angular Separation [mas]	0
Planet Radius [Earth Radius]	0
Planet Mass or Mass*sin(i) [Earth Mass]	0
Planet Density [g/cm**3]	0
Eccentricity	0
Insolation Flux [Earth Flux]	0
Equilibrium Temperature [K]	0
Inclination [deg]	0
Transit Midpoint [days]	0
Transit Depth [%]	0
Transit Duration [hours]	0
Radial Velocity Amplitude [m/s]	0
dtype: int64	
Exoplanets	
Star ID	0
Host Name	0
System ID	0
Spectral Type	0
Stellar Radius [Solar Radius]	0
Stellar Mass [Solar mass]	0
Stellar Metallicity [dex]	0
Stellar Luminosity [log(Solar)]	0
Stellar Surface Gravity [log10(cm/s**2)]	0
Stellar Age [Gyr]	0
Stellar Density [g/cm**3]	0
Stellar Rotational Velocity [km/s]	0
Stellar Rotational Period [days]	0
Systemic Radial Velocity [km/s]	0
dtype: int64	
Stars	
System ID	0
Host Name	0
Number of Stars	0
Number of Planets	0
RA [deg]	0
Dec [deg]	0
Distance [pc]	0
dtype: int64	
Solar System	
Discovery ID	0
Host Name	0
Discovery Facility	0
Discovery Year	0
Discovery Method	0
dtype: int64	
Discoveries	

```
In [36]: print(new_planet_data)
```

	Planet ID	Host Name	Planet Name	Planet Letter	Number of Moons	\
0	1	55 Cnc	55 Cnc e	e	0	
1	2	AU Mic	AU Mic b	b	0	
2	3	AU Mic	AU Mic c	c	0	
3	4	CoRoT-18	CoRoT-18 b	b	0	
4	5	G 9-40	G 9-40 b	b	0	
..	...	...	...	...	...	
157	158	WASP-96	WASP-96 b	b	0	
158	159	Wolf 327	Wolf 327 b	b	0	
159	160	XO-2 N	XO-2 N b	b	0	
160	161	XO-7	XO-7 b	b	0	
161	162	HD 39091	pi Men c	c	0	

	Orbital Period [days]	Orbit Semi-Major Axis [au]	\
0	0.736547	0.01544	
1	8.463446	0.07000	
2	18.859023	0.11900	
3	1.900069	0.02950	
4	5.745998	0.04180	
..	...	...	
157	3.425260	0.04530	
158	0.573474	0.01000	
159	2.615862	0.03680	
160	2.864142	0.04421	
161	6.267840	0.06839	

	Angular Separation [mas]	Planet Radius [Earth Radius]	\
0	1.2300	1.875000	
1	7.2000	4.790000	
2	12.2000	2.790000	
3	0.0386	14.680000	
4	1.5000	1.900000	
..	...	...	
157	0.1290	13.450000	
158	0.3500	1.240000	
159	0.2390	11.130537	
160	0.1890	15.389957	
161	3.7400	2.018900	

	Planet Mass or Mass*sin(i) [Earth Mass]	Planet Density [g/cm**3]	\
0	7.99000	6.660	
1	8.99000	0.490	
2	14.46000	3.900	
3	1102.82000	2.200	
4	4.00000	3.200	
..	...	...	
157	152.55000	0.370	
158	2.53000	7.240	
159	179.89178	0.715	
160	225.34147	0.340	
161	3.63000	2.800	

	Eccentricity	Insolation Flux [Earth Flux]	Equilibrium Temperature [K]	\
0	0.050	2657.8322	1958.0	
1	0.070	21.2000	554.8	
2	0.180	7.3000	424.7	

3	0.080	823.7009	1550.0
4	0.000	6.2700	440.6
..	...	...	...
157	0.000	517.0000	1285.0
158	0.000	233.9000	996.0
159	0.028	428.0360	1328.0
160	0.038	1540.0000	1743.0
161	0.000	309.0000	1169.8

	Inclination [deg]	Transit Midpoint [days]	Transit Depth [%] \
0	83.590	2457063.210	0.038467
1	88.390	2458330.351	0.249000
2	89.460	2458342.223	0.085000
3	86.500	2455321.724	2.571701
4	89.030	2459503.327	0.365000
..	...	...	...
157	85.600	2456258.062	1.380000
158	84.890	2459252.981	0.081839
159	88.010	2454508.738	1.336300
160	83.450	2457917.475	0.909000
161	87.045	2458425.789	0.026759

	Transit Duration [hours]	Radial Velocity Amplitude [m/s]
0	1.543900	6.020
1	3.240000	3.800
2	4.290000	4.696
3	2.387000	590.000
4	1.693000	3.220
..	...	...
157	2.426400	62.000
158	0.900451	3.540
159	2.683900	90.170
160	2.772000	80.500
161	2.952024	1.185

[162 rows x 19 columns]

```
In [37]: #merging or Linking tables Stars with Systems before MySQL Lab, one to Many
# we can use map() and dict() because system_data is already normalized
#and each row has a unique system or host name

#AUTOMATICALLY build the mapping dictionary from your stars DataFrame

#This pairs every star name with its MySQL-ready ID, no matter how many hundreds th

system_id_lookup = dict(zip(system_data['Host Name'], system_data['System ID']))

#Use .map() to fill the entire 'star_id' column at once

stars_data['System ID'] = stars_data['Host Name'].map(system_id_lookup)

#Clean up: Drop the text star_name column so MySQL stays perfectly relational

print(stars_data)
```

	Star ID	Host Name	System ID	Spectral Type	\
0	1	55 Cnc	1	G8 V	
1	2	AU Mic	2	M1 V	
3	3	CoRoT-18	3	G9 V	
4	4	G 9-40	4	M2.5 V	
5	5	GJ 1132	5	M4.5 V	
..	...	...	...	...	
157	127	WASP-96	126	G8	
158	128	Wolf 327	127	M2.5 V	
159	129	XO-2 N	128	K0 V	
160	130	XO-7	129	G0 V	
161	131	HD 39091	130	G0 V	

	Stellar Radius [Solar Radius]	Stellar Mass [Solar mass]	\
0	0.9430	0.9050	
1	0.8620	0.6350	
3	1.0000	0.9500	
4	0.3026	0.2952	
5	0.2211	0.1945	
..	...	...	
157	1.0500	1.0600	
158	0.4060	0.4050	
159	0.9900	0.9710	
160	1.4800	1.4050	
161	1.1700	1.0700	

	Stellar Metallicity [dex]	Stellar Luminosity [log(Solar)]	\
0	0.350	-0.19695	
1	0.010	-0.97757	
3	-0.100	-0.17000	
4	-0.070	-1.96019	
5	-0.170	-2.32148	
..	...	...	
157	0.140	0.03226	
158	-0.170	-1.62986	
159	0.430	-0.08823	
160	0.432	0.43132	
161	0.090	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
3	4.400	0.600	
4	4.840	9.900	
5	5.037	5.000	
..	...	...	
157	4.420	8.000	
158	4.870	4.100	
159	4.433	7.800	
160	4.246	1.180	
161	4.430	2.980	

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s]	\
0	1.521203	1.23	
1	1.750000	9.20	
3	1.350000	8.00	

4	1.620000	2.00
5	25.300000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s]
0	38.80	27.40949
1	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
5	122.30	35.07880
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

[131 rows x 14 columns]

In [38]: *#merging or Linking tables Stars and Exoplanets before MySQL Lab*
creating junction data sets for junction data tables, Many to Many

```

junction = pd.merge(
    new_planet_data,
    stars_data,
    on = 'Host Name',
    how= 'inner'
)

Star_Exoplanet_Orbits = junction[['Star ID', 'Planet ID']]

print(Star_Exoplanet_Orbits)

```

	Star ID	Planet ID
0	1	1
1	2	2
2	2	3
3	3	4
4	4	5
..	...	...
160	127	158
161	128	159
162	129	160
163	130	161
164	131	162

[165 rows x 2 columns]

In [39]: *#merging or Linking tables Exoplanet and Discoveries before MySQL Lab*
creating junction data sets for junction data tables, Many to Many

```

junction = pd.merge(
    new_planet_data,
    discoveries_data,
    on = 'Host Name',
    how= 'inner'
)

Exoplanet_Discoveries = junction[['Discovery ID', 'Planet ID']]

print(Exoplanet_Discoveries)

```

	Discovery ID	Planet ID
0	1	1
1	2	2
2	3	2
3	2	3
4	3	3
..	...	...
64	52	152
65	53	155
66	54	160
67	55	161
68	56	162

[69 rows x 2 columns]

In [40]: *#removing Host Name from all datasets where it's not needed for normalization*

```

stars_data = stars_data.drop(columns=['Host Name'])
new_planet_data = new_planet_data.drop(columns=['Host Name'])
discoveries_data = discoveries_data.drop(columns=['Host Name'])

```

In [41]: *#store cleaned datasets in new CSV files*

```

#save stars data
cleaned_stars_data = stars_data.copy()
cleaned_stars_data.to_csv('cleaned_stars_data.csv', index=False)
print(cleaned_stars_data)

#save solar system data
cleaned_system_data = system_data.copy()
cleaned_system_data.to_csv('cleaned_system_data.csv', index=False)
print(cleaned_system_data)

#save discovery data
cleaned_discoveries_data = discoveries_data.copy()
cleaned_discoveries_data.to_csv('cleaned_discoveries_data.csv', index=False)
print(cleaned_discoveries_data)

#store cleaned exoplanet data
cleaned_planet_data = new_planet_data.copy()
cleaned_planet_data.to_csv('cleaned_planet_data.csv', index=False)
print(cleaned_planet_data)

#junction table Star_Exoplanet_Orbits

```

```
Star_Exoplanet_Orbits_cleaned = Star_Exoplanet_Orbits.copy()
Star_Exoplanet_Orbits_cleaned.to_csv('Star_Exoplanet_Orbits_cleaned.csv', index=False)
print(Star_Exoplanet_Orbits_cleaned)

#junction table Exoplanet Discoveries
Exoplanet_Discoveries_cleaned = Exoplanet_Discoveries.copy()
Exoplanet_Discoveries_cleaned.to_csv('Exoplanet_Discoveries_cleaned.csv', index=False)
print(Exoplanet_Discoveries_cleaned)
```

	Star ID	System ID	Spectral Type	Stellar Radius [Solar Radius]	\
0	1	1	G8 V	0.9430	
1	2	2	M1 V	0.8620	
3	3	3	G9 V	1.0000	
4	4	4	M2.5 V	0.3026	
5	5	5	M4.5 V	0.2211	
..	...	...	...	...	
157	127	126	G8	1.0500	
158	128	127	M2.5 V	0.4060	
159	129	128	K0 V	0.9900	
160	130	129	G0 V	1.4800	
161	131	130	G0 V	1.1700	

	Stellar Mass [Solar mass]	Stellar Metallicity [dex]	\
0	0.9050	0.350	
1	0.6350	0.010	
3	0.9500	-0.100	
4	0.2952	-0.070	
5	0.1945	-0.170	
..	...	...	
157	1.0600	0.140	
158	0.4050	-0.170	
159	0.9710	0.430	
160	1.4050	0.432	
161	1.0700	0.090	

	Stellar Luminosity [log(Solar)]	\
0	-0.19695	
1	-0.97757	
3	-0.17000	
4	-1.96019	
5	-2.32148	
..	...	
157	0.03226	
158	-1.62986	
159	-0.08823	
160	0.43132	
161	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
3	4.400	0.600	
4	4.840	9.900	
5	5.037	5.000	
..	...	...	
157	4.420	8.000	
158	4.870	4.100	
159	4.433	7.800	
160	4.246	1.180	
161	4.430	2.980	

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s]	\
0	1.521203	1.23	
1	1.750000	9.20	
3	1.350000	8.00	

4	1.620000	2.00
5	25.300000	2.00
..	...	...
157	1.299000	1.50
158	8.540000	2.00
159	1.410000	1.07
160	0.451084	6.00
161	0.944584	3.16

	Stellar Rotational Period [days]	Systemic Radial Velocity [km/s]
0	38.80	27.40949
1	4.86	-6.31031
3	5.40	29.55000
4	30.00	14.65000
5	122.30	35.07880
..	...	...
157	20.16	-1.23300
158	44.40	9.39000
159	28.60	46.93290
160	3.62	-12.98300
161	18.30	10.70916

[131 rows x 13 columns]

	System ID	Host Name	Number of Stars	Number of Planets	RA [deg]	\
0	1	55 Cnc	2	7	133.146837	
1	2	AU Mic	1	4	311.291137	
3	3	CoRoT-18	1	1	98.172417	
4	4	G 9-40	1	1	134.718851	
5	5	GJ 1132	1	2	153.709070	
..	...	...	...	...	...	...
157	126	WASP-96	1	1	1.046570	
158	127	Wolf 327	1	1	148.378320	
159	128	XO-2 N	2	3	117.026769	
160	129	XO-7	1	1	277.478075	
161	130	HD 39091	1	3	84.299280	

	Dec [deg]	Distance [pc]
0	28.329815	12.5855
1	-31.342450	9.7221
3	-0.031605	764.8890
4	21.074796	27.9276
5	-47.154936	12.6130
..	...	...
157	-47.360635	352.4640
158	35.570548	28.5321
159	50.225147	154.2730
160	85.233321	234.1490
161	-80.464604	18.2702

[130 rows x 7 columns]

	Discovery ID	Discovery Facility	\
0	1	McDonald Observatory	
1	2	Transiting Exoplanet Survey Satellite (TESS)	
2	3	Transiting Exoplanet Survey Satellite (TESS)	
3	4	CoRoT	
4	5	K2	

5	6	Cerro Tololo Inter-American Observatory
6	7	MEarth Project
7	8	Transiting Exoplanet Survey Satellite (TESS)
8	9	W. M. Keck Observatory
9	10	Calar Alto Observatory
10	11	K2
13	12	Transiting Exoplanet Survey Satellite (TESS)
14	13	HATNet
15	14	HATNet
19	15	HATNet
20	16	HATSouth
21	17	Multiple Observatories
24	18	Haute-Provence Observatory
25	19	La Silla Observatory
28	20	Transiting Exoplanet Survey Satellite (TESS)
30	21	Subaru Telescope
34	22	W. M. Keck Observatory
38	23	K2
42	24	K2
56	25	K2
58	26	K2
66	27	KELT-North
67	28	Kepler
69	29	Kepler
70	30	Kepler
74	31	Kepler
75	32	Kepler
85	33	Qatar
89	34	Qatar
101	35	Transiting Exoplanet Survey Satellite (TESS)
105	36	Transiting Exoplanet Survey Satellite (TESS)
127	37	SuperWASP
128	38	SuperWASP-South
129	39	SuperWASP-North
130	40	SuperWASP
132	41	SuperWASP-South
135	42	SuperWASP-North
136	43	WASP-South
137	44	SuperWASP-South
139	45	SuperWASP
141	46	SuperWASP
142	47	SuperWASP
143	48	SuperWASP
144	49	SuperWASP
145	50	SuperWASP
147	51	SuperWASP
151	52	SuperWASP
154	53	SuperWASP
159	54	XO
160	55	XO
161	56	Transiting Exoplanet Survey Satellite (TESS)

	Discovery Year	Discovery Method
0	2004	Radial Velocity
1	2020	Transit
2	2021	Transit

3	2011	Transit
4	2019	Transit
5	2015	Transit
6	2009	Transit
7	2022	Transit
8	2004	Radial Velocity
9	2021	Radial Velocity
10	2017	Transit
13	2024	Transit
14	2008	Transit
15	2010	Transit
19	2007	Transit
20	2016	Transit
21	2016	Transit
24	2005	Radial Velocity
25	2019	Radial Velocity
28	2019	Transit
30	2007	Radial Velocity
34	1999	Radial Velocity
38	2016	Transit
42	2024	Transit
56	2018	Transit
58	2015	Transit
66	2017	Transit
67	2014	Transit
69	2011	Transit
70	2012	Transit
74	2016	Transit
75	2013	Transit
85	2017	Transit
89	2019	Transit
101	2023	Transit
105	2025	Transit
127	2008	Transit
128	2017	Transit
129	2016	Transit
130	2016	Transit
132	2016	Transit
135	2020	Transit
136	2019	Transit
137	2018	Transit
139	2019	Transit
141	2020	Transit
142	2009	Transit
143	2010	Transit
144	2007	Transit
145	2011	Transit
147	2012	Transit
151	2014	Transit
154	2015	Transit
159	2007	Transit
160	2019	Transit
161	2018	Transit

	Planet ID	Planet Name	Planet Letter	Number of Moons	\
0	1	55 Cnc e	e	0	
1	2	AU Mic b	b	0	

2	3	AU Mic c	c	0
3	4	CoRoT-18 b	b	0
4	5	G 9-40 b	b	0
..	...	...	...	...
157	158	WASP-96 b	b	0
158	159	Wolf 327 b	b	0
159	160	XO-2 N b	b	0
160	161	XO-7 b	b	0
161	162	pi Men c	c	0

	Orbital Period [days]	Orbit Semi-Major Axis [au]	\
0	0.736547	0.01544	
1	8.463446	0.07000	
2	18.859023	0.11900	
3	1.900069	0.02950	
4	5.745998	0.04180	
..	...	...	
157	3.425260	0.04530	
158	0.573474	0.01000	
159	2.615862	0.03680	
160	2.864142	0.04421	
161	6.267840	0.06839	

	Angular Separation [mas]	Planet Radius [Earth Radius]	\
0	1.2300	1.875000	
1	7.2000	4.790000	
2	12.2000	2.790000	
3	0.0386	14.680000	
4	1.5000	1.900000	
..	...	...	
157	0.1290	13.450000	
158	0.3500	1.240000	
159	0.2390	11.130537	
160	0.1890	15.389957	
161	3.7400	2.018900	

	Planet Mass or Mass*sin(i) [Earth Mass]	Planet Density [g/cm**3]	\
0	7.99000	6.660	
1	8.99000	0.490	
2	14.46000	3.900	
3	1102.82000	2.200	
4	4.00000	3.200	
..	...	...	
157	152.55000	0.370	
158	2.53000	7.240	
159	179.89178	0.715	
160	225.34147	0.340	
161	3.63000	2.800	

	Eccentricity	Insolation Flux [Earth Flux]	Equilibrium Temperature [K]	\
0	0.050	2657.8322	1958.0	
1	0.070	21.2000	554.8	
2	0.180	7.3000	424.7	
3	0.080	823.7009	1550.0	
4	0.000	6.2700	440.6	
..	...	...	...	

157	0.000	517.0000	1285.0
158	0.000	233.9000	996.0
159	0.028	428.0360	1328.0
160	0.038	1540.0000	1743.0
161	0.000	309.0000	1169.8

	Inclination [deg]	Transit Midpoint [days]	Transit Depth [%]	\
0	83.590	2457063.210	0.038467	
1	88.390	2458330.351	0.249000	
2	89.460	2458342.223	0.085000	
3	86.500	2455321.724	2.571701	
4	89.030	2459503.327	0.365000	
..	...	...	...	
157	85.600	2456258.062	1.380000	
158	84.890	2459252.981	0.081839	
159	88.010	2454508.738	1.336300	
160	83.450	2457917.475	0.909000	
161	87.045	2458425.789	0.026759	

	Transit Duration [hours]	Radial Velocity Amplitude [m/s]
0	1.543900	6.020
1	3.240000	3.800
2	4.290000	4.696
3	2.387000	590.000
4	1.693000	3.220
..	...	...
157	2.426400	62.000
158	0.900451	3.540
159	2.683900	90.170
160	2.772000	80.500
161	2.952024	1.185

[162 rows x 18 columns]

	Star ID	Planet ID
0	1	1
1	2	2
2	2	3
3	3	4
4	4	5
..	...	...
160	127	158
161	128	159
162	129	160
163	130	161
164	131	162

[165 rows x 2 columns]

	Discovery ID	Planet ID
0	1	1
1	2	2
2	3	2
3	2	3
4	3	3
..	...	...
64	52	152
65	53	155

66	54	160
67	55	161
68	56	162

[69 rows x 2 columns]

In []: *# my code ends here*

In []: